

Practice Problems - Set B

Answer Key

This comprehensive practice set covers Chapters 16-18 of Feenstra & Taylor. Work through all questions to prepare for Midterm 2.

1. (12 points) BOP Transactions and Accounting

Show how each of the following affects the U.S. balance of payments. For each transaction, identify the debit and credit items and the specific accounts affected.

- (a) The U.S. government forgives \$100 million in debt owed by a developing country.

Description	BOP account	Account detail	Credit/Debit
Debt forgiveness (transfer out)	KA ↓	$KA^{out} \uparrow$	-\$100M
Decrease in foreign assets owned by US	FA ↑	$EX^A \uparrow$	+\$100M

Debt forgiveness is a unilateral transfer recorded in the capital account (debit). The off-setting entry is the write-off of the loan asset (credit in financial account). The two entries sum to zero, maintaining BOP balance.

- (b) A Japanese car manufacturer builds a \$200 million factory in Tennessee, paying from its bank account in New York.

Description	BOP account	Account detail	Credit/Debit
Japanese FDI into US	FA ↑	$EX^A \uparrow$	+\$200M
Decrease in Japanese deposits in US	FA ↓	$IM^A \downarrow$	-\$200M

The factory is a U.S. asset acquired by foreigners (credit in FA). Payment from the Japanese company's U.S. bank account reduces foreign-owned assets in the U.S. (debit in FA).

- (c) A U.S. mutual fund purchases \$50 million in Australian stocks, financing the purchase by borrowing from an Australian bank.

Description	BOP account	Account detail	Credit/Debit
Purchase of Australian stocks	FA ↓	$IM^A \uparrow$	-\$50M
Loan from Australian bank (US liability)	FA ↑	$EX^A \uparrow$	+\$50M

Buying Australian stocks is a U.S. acquisition of foreign assets (debit in FA). The loan creates a U.S. liability to Australia (credit in FA). Both are financial account transactions.

2. (18 points) Long-Run Budget Constraint

Consider an economy that exists for three periods ($t = 0, 1, 2$) with no government, investment, or unilateral transfers ($G = I = NUT = 0$). The world interest rate is $r^* = 0.08$.

The country has initial external wealth $W_{-1} = -200$ and the following trade balances:

- $TB_0 = 50$
- $TB_1 = 40$
- $TB_2 = ?$ (to be determined)

(a) Write the period-by-period budget constraints for periods 0 and 1.

Period 0:

$$W_0 = TB_0 + (1 + r^*)W_{-1} = 50 + (1.08)(-200) = 50 - 216 = -166$$

Period 1:

$$W_1 = TB_1 + (1 + r^*)W_0 = 40 + (1.08)(-166) = 40 - 179.28 = -139.28$$

(b) If the country must have zero external wealth at the end of period 2 ($W_2 = 0$), what must TB_2 be?

For $W_2 = 0$:

$$0 = TB_2 + (1 + r^*)W_1$$

$$TB_2 = -(1.08)(-139.28) = 150.42$$

The country must run a trade surplus of 150.42 in the final period to pay off its net debtor position and the accumulated interest.

(c) Derive the long-run budget constraint by expressing the present value of wealth in terms of trade balances.

From $W_2 = 0$:

$$0 = TB_2 + (1 + r^*)W_1$$

$$W_1 = -\frac{TB_2}{1 + r^*}$$

Similarly:

$$W_1 = TB_1 + (1 + r^*)W_0$$

$$W_0 = \frac{W_1 - TB_1}{1 + r^*} = -\frac{TB_2}{(1 + r^*)^2} - \frac{TB_1}{1 + r^*}$$

And:

$$W_0 = TB_0 + (1 + r^*)W_{-1}$$

Combining:

$$-(1 + r^*)W_{-1} = TB_0 + \frac{TB_1}{1 + r^*} + \frac{TB_2}{(1 + r^*)^2}$$

Long-run budget constraint:

$$-(1 + r^*)W_{-1} = \sum_{t=0}^2 \frac{TB_t}{(1 + r^*)^t}$$

The present value of trade balances must equal the negative of the present value of initial wealth (initial debt plus interest).

(d) Verify that the LRBC holds with your calculated TB_2 .

$$\text{LHS: } -(1.08)(-200) = 216$$

RHS:

$$\begin{aligned} TB_0 + \frac{TB_1}{1.08} + \frac{TB_2}{(1.08)^2} &= 50 + \frac{40}{1.08} + \frac{150.42}{1.1664} \\ &= 50 + 37.04 + 128.96 = 216 \end{aligned}$$

$$216 = 216 \text{ (checks out)} \rightarrow \text{the LRBC holds.}$$

3. (12 points) Consumption Smoothing with Different Preferences

Consider a two-period economy with $r^* = 0.10$ and $W_{-1} = 0$. Income is $Q_0 = 100$, $Q_1 = 132$. Compare consumption and trade balances under two different utility functions:

Case A: $U = \ln(C_0) + \ln(C_1)$ (log utility)

Case B: $U = \min(C_0, C_1)$ (perfect complements)

Hint: The optimality condition in the log case is given by $C_1 = (1 + r^*)C_0$. This condition is also known as an *Euler Equation* and equates the marginal utility of consumption today with the one obtained by saving and consuming the next period.

(a) For each case, find optimal consumption in both periods.

$$\text{LRBC: } C_0 + \frac{C_1}{1.10} = 100 + \frac{132}{1.10} = 100 + 120 = 220$$

Case A (log utility):

$$\text{Optimal condition: } C_1 = (1 + r^*)C_0 = 1.10C_0$$

Substituting into LRBC:

$$C_0 + \frac{1.10C_0}{1.10} = 220$$

$$2C_0 = 220$$

$$C_0 = 110, \quad C_1 = 121$$

Case B (perfect complements):

Optimal condition: $C_0 = C_1 = C$

Substituting into LRBC:

$$C + \frac{C}{1.10} = 220$$

$$C \left(\frac{2.10}{1.10} \right) = 220$$

$$C = 220 \times \frac{1.10}{2.10} = 115.24$$

$$C_0 = C_1 = 115.24$$

(b) Calculate the trade balance in period 0 for each case. Which case involves more borrowing?

Case A:

$$TB_0 = Q_0 - C_0 = 100 - 110 = -10$$

Case B:

$$TB_0 = Q_0 - C_0 = 100 - 115.24 = -15.24$$

Case B involves more borrowing (larger trade deficit in period 0): With perfect complements, the household values consumption smoothing more highly and is willing to borrow more to achieve perfectly equal consumption across periods, despite the interest cost.

With log utility, the household is willing to tolerate some consumption growth (from 110 to 121) to take advantage of the positive interest rate.

4. (15 points) Gains from Financial Globalization

Country A is considering two investment projects. It can access world capital markets at $r^* = 0.06$. The household has utility $U = \ln(C_0) + \ln(C_1)$ and initial wealth $W_{-1} = 0$.

Baseline: $Q_0 = 500, Q_1 = 500$

Project 1: Invest $I_0^1 = 100$, get $Q_1 = 615$

Project 2: Invest $I_0^2 = 80$, get $Q_1 = 592$

- (a) What is the return for each investment (also known as its marginal product)? Compare them to the world interest rate.

Project 1:

$$MPK_1 = \frac{\Delta Q_1}{I_0^1} = \frac{615 - 500}{100} = \frac{115}{100} = 1.15 \text{ or } 115\%$$

Project 2:

$$MPK_2 = \frac{\Delta Q_1}{I_0^2} = \frac{592 - 500}{80} = \frac{92}{80} = 1.15 \text{ or } 115\%$$

Both projects have the same MPK of 115%, which is much higher than $r^* = 6\%$.

- (b) Should the country undertake one project, both projects, or neither? Calculate utility in each case to justify your answer.

Hint: The optimality condition in the log case is given by $C_1 = (1 + r^*)C_0$. This condition is also known as an *Euler Equation* and equates the marginal utility of consumption today with the one obtained by saving and consuming the next period.

Since both projects have $MPK = 115\% > r^* = 6\%$, both are profitable.

For log utility we can replace the hint in the LRBC and get: $C_0 = \frac{1}{2} \left[Q_0 + \frac{Q_1}{1.06} - I_0 \right]$

No investment:

$$C_0 = \frac{1}{2} \left[500 + \frac{500}{1.06} \right] = \frac{1}{2} [500 + 471.70] = 485.85$$

$$C_1 = 1.06 \times 485.85 = 515$$

$$U = \ln(485.85) + \ln(515) = 6.186 + 6.244 = 12.43$$

Project 1 only:

$$C_0 = \frac{1}{2} \left[500 + \frac{615}{1.06} - 100 \right] = \frac{1}{2} [500 + 580.19 - 100] = 490.09$$

$$C_1 = 519.5$$

$$U = \ln(490.09) + \ln(519.5) = 6.194 + 6.253 = 12.447$$

Project 2 only:

$$C_0 = \frac{1}{2} \left[500 + \frac{592}{1.06} - 80 \right] = \frac{1}{2} [500 + 558.49 - 80] = 489.25$$

$$C_1 = 518.61$$

$$U = \ln(489.25) + \ln(518.61) = 6.193 + 6.251 = 12.444$$

Both projects:

$$C_0 = \frac{1}{2} \left[500 + \frac{707}{1.06} - 180 \right] = \frac{1}{2} [500 + 667.0 - 180] = 493.49$$

$$C_1 = 523.10$$

$$U = \ln(493.49) + \ln(523.10) = 6.201 + 6.260 = 12.461$$

Conclusion: Undertake both projects. This gives the highest utility (12.461).

- (c) What would happen if the country did not have access to international capital markets? Would it still invest in both projects?

Without access to capital markets (autarky), the country cannot borrow or lend. It must consume its GDP each period: $C_0 = Q_0 - I_0$ and $C_1 = Q_1$.

Since the household wants smooth consumption but cannot borrow/lend, large investment reduces C_0 substantially, creating volatile consumption.

The country would likely invest less or not at all in autarky because:

- Investment of 180 reduces C_0 to 320 (from 500)
- C_1 rises to 707
- This extreme volatility (320 vs. 707) is very costly with log utility

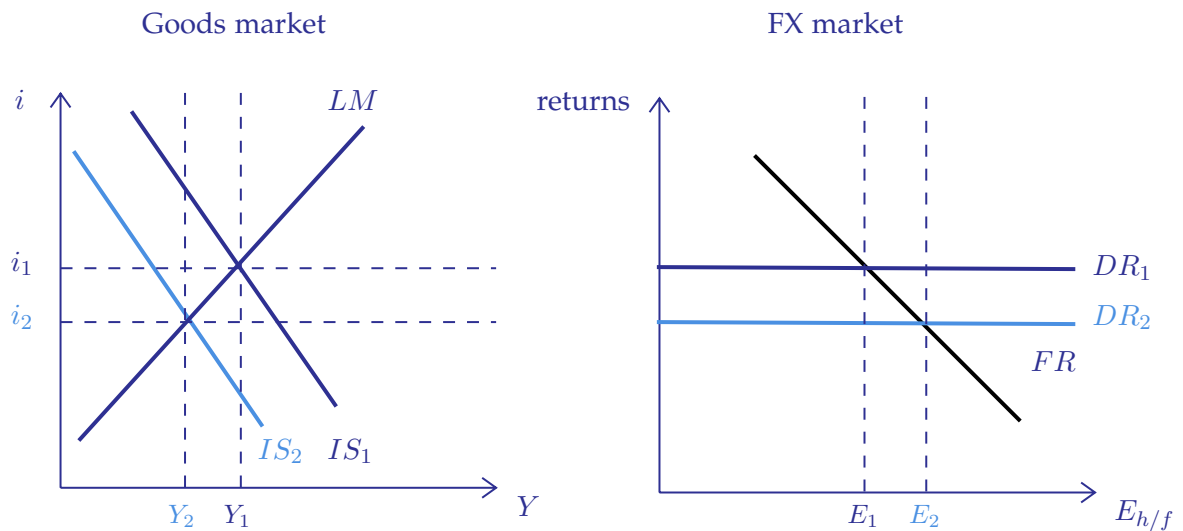
Financial globalization allows the country to invest optimally (both projects) while maintaining smooth consumption (493.49 vs. 523.10) by borrowing in period 0. This illustrates the key gain from financial globalization: decoupling investment from saving decisions.

5. (12 points) IS-LM-FX: Domestic Demand Shock

A country experiences a drop in consumer confidence, causing autonomous consumption to fall. The country operates under a floating exchange rate.

- (a) Analyze the effects using IS-LM-FX diagrams. State the impact on Y , i , $E_{h/f}$, C , I , and TB .

A fall in autonomous consumption shifts the IS curve left.



Effects:

- $Y \downarrow$ (output falls)
- $i \downarrow$ (interest rate falls, movement along LM)
- $E_{h/f} \uparrow$ (currency depreciates as DR shifts down)
- $C \downarrow$ (consumption falls—both from the autonomous drop and from lower income)
- $I \uparrow$ (investment increases due to lower interest rate)
- TB : Ambiguous
 - Positive effect: Depreciation improves competitiveness
 - Negative effect: Lower income reduces imports (improves TB)
 - The income effect typically dominates, so $TB \uparrow$

The automatic depreciation provides some stabilization, but output still falls. The increase in investment and trade balance partially offsets the consumption decline.

(b) What policy could the government use to stabilize output at its original level? Would fiscal or monetary policy be more appropriate?

Monetary policy (increase money supply):

- Shift LM right
- Further lower interest rates
- Further depreciate currency
- Stimulate investment and trade balance

- Offset the output decline

Fiscal policy (increase government spending or lower taxes):

- Shift IS right (back toward original position)
- Directly offset the fall in consumption
- But raises interest rates and appreciates currency
- Crowds out investment and trade balance

More appropriate: Monetary policy. A monetary expansion works with the market forces (depreciation) rather than against them. It avoids crowding out and may be faster to implement than fiscal policy. However, if interest rates are already near zero (zero lower bound), fiscal policy becomes necessary.

6. (15 points) IS-LM-FX: Policy Comparison and Supply Shocks

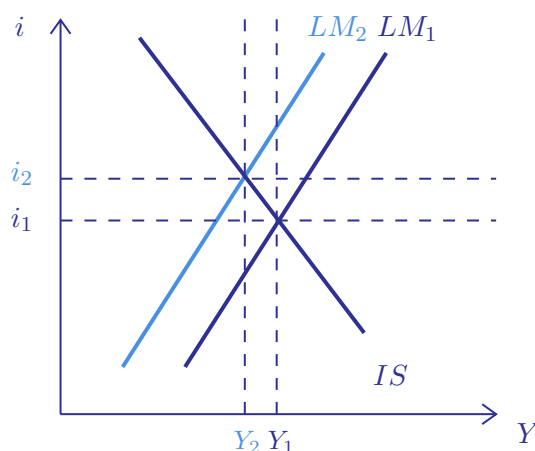
Consider a small open economy initially in equilibrium. The economy experiences a negative supply shock (oil price increase) that reduces potential output and increases the domestic price level.

- (a) How does this supply shock affect the LM curve? Draw an IS-LM diagram showing the initial effect on output and interest rates.

A negative supply shock increases the domestic price level P . Since the LM curve depends on real money balances M/P , an increase in P (with unchanged nominal money supply M) reduces M/P .

This causes the LM curve to shift **left**.

Goods market



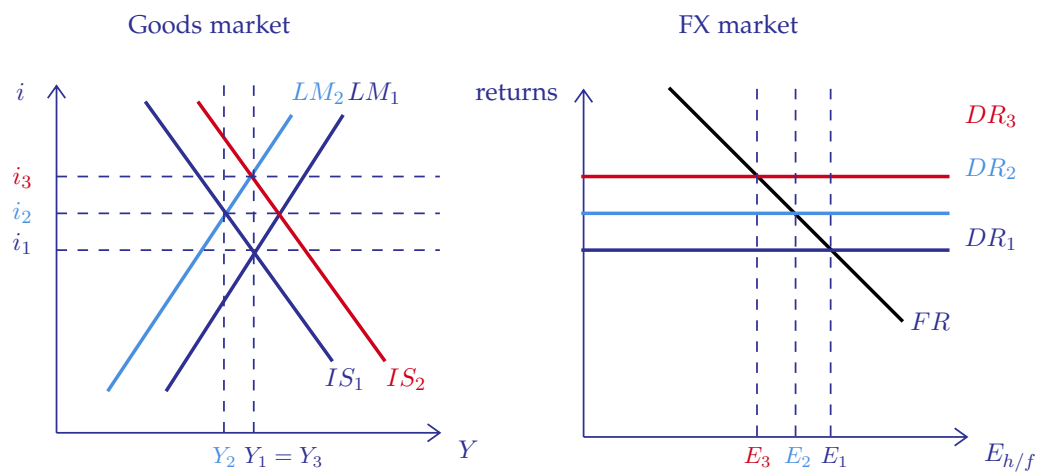
Initial effects:

- LM shifts left from LM_1 to LM_2 (due to lower real money balances $M/P \downarrow$)
- Output falls: $Y_1 \rightarrow Y_2$, so $Y \downarrow$
- Interest rate rises: $i_1 \rightarrow i_2$, so $i \uparrow$

The supply shock is **stagflationary**: it reduces output and increases both the price level and the interest rate.

- (b) The government wants to stabilize output at its original level using fiscal policy (increasing G or lowering T). Under a **floating exchange rate**, show the combined effects of the supply shock and fiscal response. What happens to Y , i , $E_{h/f}$, and TB ?

To stabilize output, the government increases G , which shifts the IS curve right.



Step 1 - Supply shock: LM shifts left (blue), moving from point 1 to point 2: $Y \downarrow$ to Y_2 , $i \uparrow$ to i_2 , $E_{h/f} \downarrow$ to E_2 (appreciation)

Step 2 - Fiscal response: Government increases G , shifting IS right (red) from IS_1 to IS_2 . This moves the economy from point 2 to point 3: output returns to Y_1 ($Y_3 = Y_1$), interest rate rises further to i_3 , currency appreciates further to E_3

Final effects (comparing point 1 to point 3):

- Y : unchanged ($Y_3 = Y_1$) — fiscal policy successfully stabilizes output
- $i \uparrow\uparrow$: much higher ($i_3 > i_2 > i_1$)
- $E_{h/f} \downarrow\downarrow$: strong appreciation ($E_3 < E_2 < E_1$)
- $TB \downarrow$: trade balance worsens significantly due to appreciation

Policy assessment: While fiscal policy achieves the output goal, it comes at a cost: very high interest rates (crowding out investment) and sharp currency appreciation (hurting exporters). Under floating rates, stabilization policy creates winners (government con-

tractors, import-competing sectors) and losers (exporters, borrowers).

- (c) Now suppose the economy operates under a **fixed exchange rate** instead. Can the government successfully stabilize output using fiscal policy? Explain what additional action the central bank must take and whether this helps or hinders the fiscal policy objective.

Under a fixed exchange rate, fiscal policy works even better for output stabilization.

After the supply shock shifts LM left, the government increases G (IS shifts right). The higher interest rate from both the supply shock and fiscal expansion would normally cause the currency to appreciate.

Central bank action: To maintain the fixed exchange rate $\bar{E}$, the central bank must prevent appreciation by buying foreign currency (selling domestic currency). This increases the money supply, shifting the LM back to the right, partially or fully offsetting the initial supply shock. (in a nutshell, moving M to undo the effect of the higher P on M/P .)

Does this help or hinder fiscal policy?: This **helps** the fiscal policy objective! The monetary accommodation:

- Prevents interest rates from rising as much (less crowding out of investment)
- Prevents currency appreciation (no deterioration in trade balance)
- Amplifies the output effect of fiscal policy

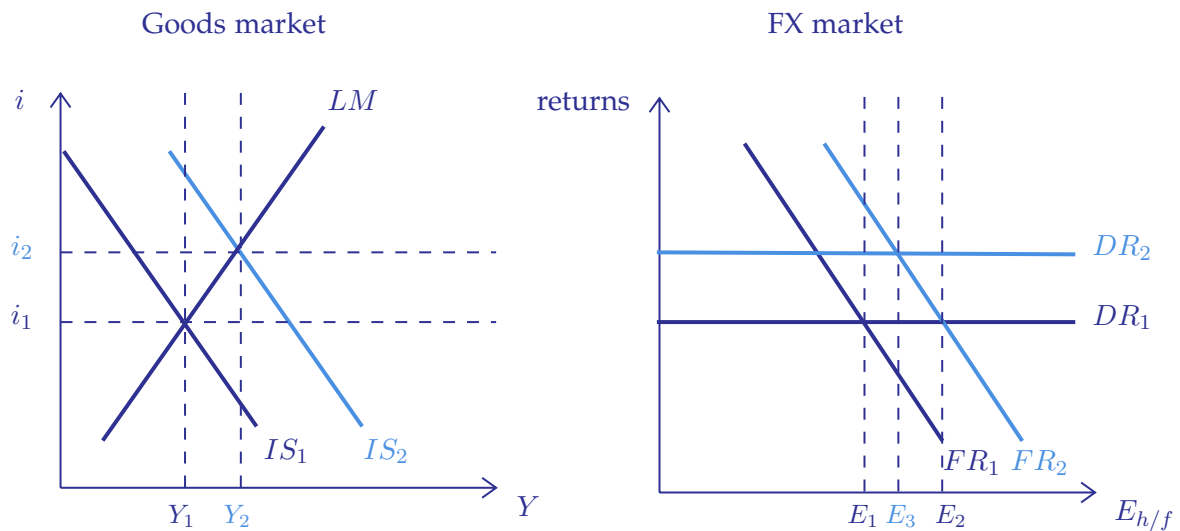
Final result: Output can return to Y_1 (or even exceed it if the central bank fully reverses the LM shift) with smaller increases in interest rates and no appreciation. Fiscal policy is more effective under fixed rates because monetary policy must accommodate it.

7. (15 points) IS-LM-FX: Comparing Exchange Rate Regimes

Consider an economy that experiences an increase in foreign interest rates ($i^* \uparrow$). Analyze the effects under both floating and fixed exchange rate regimes.

- (a) Under a floating exchange rate, what happens to Y , i , and $E_{h/f}$? Draw diagrams.

When $i^* \uparrow$, the FR curve shifts right (foreign assets more attractive).



Initial shock in FX market:

- FR shifts right ($FR = i^* + \left(\frac{E_{h/f}^e}{E_{h/f}} - 1\right)$)
- Home currency depreciates: $E_{h/f} \uparrow$

Effect in goods market:

- Depreciation improves competitiveness $\rightarrow$ TB increases $\rightarrow$ IS shifts right
- Output increases: $Y \uparrow$
- Interest rate increases: $i \uparrow$ (movement along LM)

The increase in the interest rate prompts an additional shift in the DR curve in the FX market diagram. In the final equilibrium both FR and DR have shifted and the final exchange rate is depreciated relative to the initial level. In this case, the home country benefits from the foreign rate increase through automatic depreciation that boosts exports.

(b) Under a fixed exchange rate, what must the central bank do? What are the effects on output?

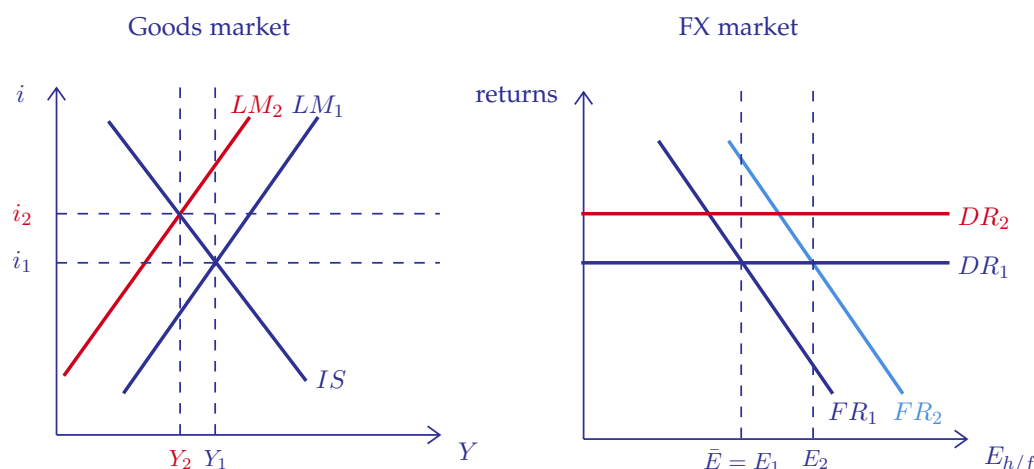
Under a fixed exchange rate regime when $i^* \uparrow$:

FX market: The higher foreign interest rate shifts the FR curve right, creating depreciation pressure on the home currency (pushing $E_{h/f}$ from E_1 toward E_2).

Central bank action to maintain the peg $E = \bar{E} = E_1$: The central bank must resist depreciation by selling foreign reserves and buying domestic currency. This contracts the money supply ($M \downarrow$), shifting the LM curve **left**. The higher domestic interest rate that results restores UIP at the original exchange rate: home assets become attractive enough to offset the higher i^* , eliminating depreciation pressure.

Key difference from the floating case: Under a credible fixed rate, the exchange rate never

actually moves. Therefore, there is **no depreciation**, no improvement in competitiveness (no increase in TB), and **no shift in the IS curve**. The IS curve stays put throughout.



Final equilibrium:

- $Y \downarrow$: Output falls. The LM shifts left with no offsetting IS shift, so the new goods-market equilibrium is at lower output.
- $i \uparrow$: The interest rate rises to i_2 to satisfy UIP at the fixed exchange rate.
- $E_{h/f} = \bar{E}$: The exchange rate is successfully held at E_1 .

Contrast with the floating rate case (Part A):

- Under *floating*: depreciation occurs automatically, IS shifts right, and output *expands* ($Y \uparrow$). The foreign rate shock is in fact *expansionary*.
- Under *fixed*: no depreciation occurs, IS is unchanged, and the CB must contract money supply to defend the peg. The same shock is *contractionary* ($Y \downarrow$).

This contrast illustrates a central cost of fixed exchange rate regimes: the central bank loses monetary policy autonomy. External shocks that require tightening to defend the peg are transmitted directly as output contractions, with no exchange-rate buffer.

8. (10 points) Small Open Economy Assumptions

- (a) Explain what it means for a country to be a "small open economy" in the IS-LM-FX framework. What are the key assumptions?

A small open economy is characterized by:

1. **Price taker in world markets:** Cannot influence world interest rate, foreign output, or the foreign prices level: i^* , Y^* , and P^* can be thought as exogenous.
2. **Perfect capital mobility:** Domestic and foreign assets are perfect substitutes, Capital

flows freely across borders, and as a consequence, the UIP holds:

$$i = i^* + \frac{E^e - E}{E}$$

Implications:

- The country faces a perfectly elastic supply of foreign capital at the world rate
- Domestic policies affect domestic variables but not world variables
- The position of the FR curve is determined solely by exogenous variables (i^* , E^e , which the small economy cannot influence), but the curve itself slopes downward in $E_{h/f}$, and the equilibrium exchange rate is determined endogenously by the UIP.

Most countries except the U.S., EU, Japan, and China can be approximated as small open economies.

- (b) If a country's bond market is large enough to affect world interest rates, how would the analysis change?

For a large open economy (e.g., United States):

Key difference: Domestic policy affects i^*

Example: U.S. monetary expansion

- LM shifts right $\rightarrow i$ falls
- This also lowers i^* (spillover effect)
- FR shifts down (lower i^*)
- Dollar depreciates more than in small economy case, other countries' currencies appreciate, and output effects spread globally.

Policy implications: Large economies must consider international spillovers. This can have additional implications about the potential benefits of coordination efforts by major central banks and the presence of "beggar-thy-neighbor" incentives by large economies (i.e., the temptation to influence global prices and quantities —e.g., the terms of trade or real exchange rates— in their favor and in detriment of other economies)

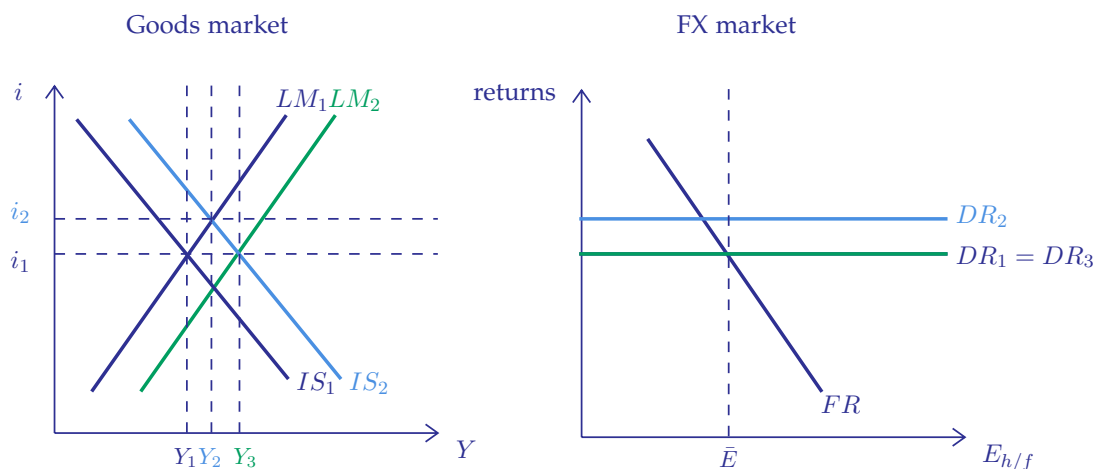
in this case, the IS-LM-FX model becomes more complex because foreign variables become endogenous.

9. (15 points) IS-LM-FX: Comprehensive Policy Analysis

An economy is initially in equilibrium with $Y = Y^*$ (full employment). The government wants to increase output above full employment through expansionary fiscal policy ($G \uparrow$) while maintaining a fixed exchange rate.

- (a) What must the central bank do to maintain the fixed exchange rate after the fiscal expansion? Use an IS-LM-FX diagram to illustrate.

Fiscal expansion ($G \uparrow$) shifts IS right. The higher output and interest rate make domestic assets more attractive, generating capital inflows and appreciation pressure on the home currency ($E_{h/f}$ would fall). To prevent appreciation and maintain $\bar{E}$, the central bank must buy foreign currency and sell domestic currency, expanding the money supply ($M \uparrow$) and shifting LM right. The interest rate returns toward its original level.



Sequence of events:

1. IS shifts right ($IS_1 \rightarrow IS_2$) (blue). Output rises to Y_2 , interest rate rises to i_2 , appreciation pressure builds (DR shifts up to DR_2 , pushing $E_{h/f}$ below $\bar{E}$).
2. CB buys foreign currency, expands money supply. LM shifts right ($LM_1 \rightarrow LM_2$) (green), interest rate returns to i_1 , DR returns to DR_1 , exchange rate stays at $\bar{E}$. Output rises further to Y_3 .

Conclusion: The central bank must conduct an accommodating monetary expansion. Under a fixed rate, fiscal policy automatically triggers monetary accommodation, amplifying the output effect.

- (b) What is the final effect on output compared to the same fiscal expansion under a floating exchange rate? Use IS-LM-FX diagrams to compare both regimes.

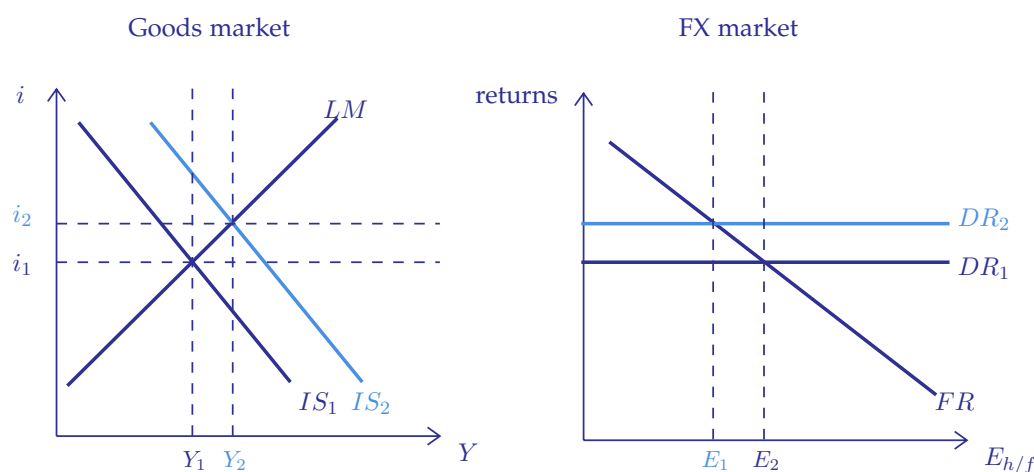
Fixed exchange rate (same as in part a):

IS shifts right and LM shifts right (CB accommodation). The interest rate returns to i_1 , the exchange rate stays at $\bar{E}$, and output rises substantially to Y_3 . There is no crowding out of investment (interest rate unchanged) or the trade balance (exchange rate unchanged).

Floating exchange rate:

IS shifts right. The higher interest rate causes the currency to appreciate ($E_{h/f} \downarrow$), which

worsens the trade balance and partially shifts IS back left. LM is unchanged throughout.



Under floating, the higher interest rate causes appreciation ($E_{h/f}$ falls from E_2 to E_1), worsening the trade balance and crowding out investment. The output gain $Y_2 - Y_1$ is smaller than under the fixed rate.

Summary:	Fixed rate	Floating rate
IS	shifts right	shifts right
LM	shifts right (CB)	unchanged
Output gain	large ($Y_3 - Y_1$)	small ($Y_2 - Y_1$)
Interest rate	unchanged (i_1)	rises ($i_2 > i_1$)
Exchange rate	unchanged ($\bar{E}$)	appreciates ($E \downarrow$)
Investment	unchanged	crowded out
Trade balance	unchanged	deteriorates

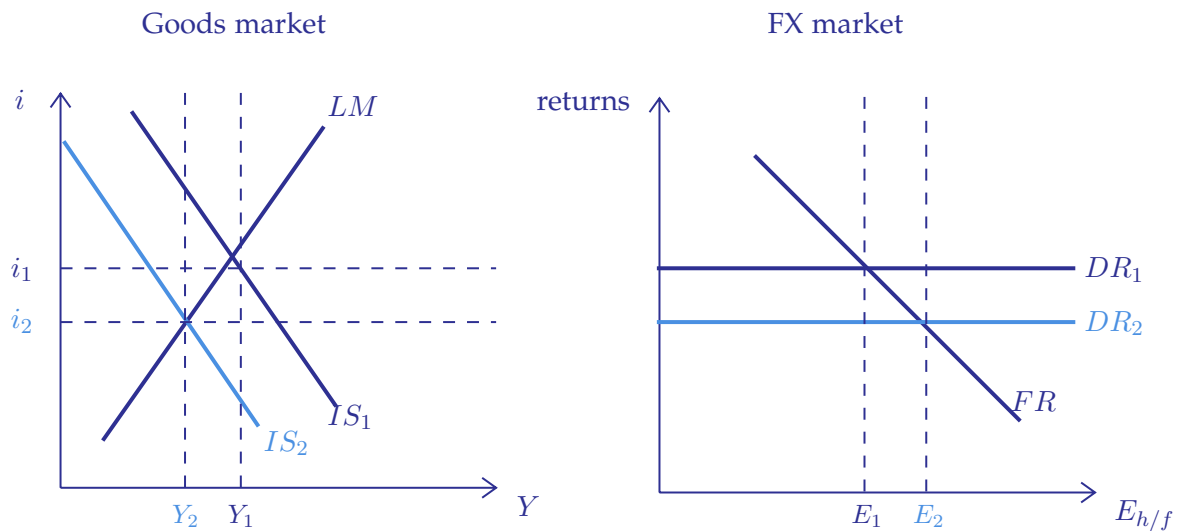
Conclusion: Fiscal policy is more effective under a fixed exchange rate because monetary accommodation eliminates crowding out of both investment and the trade balance.

10. (12 points) IS-LM-FX: Investment Shock and Policy Response

Suppose U.S. firms become pessimistic about future profitability and cut investment expenditure autonomously (i.e., independently of the interest rate). The economy operates under a **floating exchange rate** and the government makes **no policy response** in part (a).

- (a) How does the investment shock affect the IS curve? Using an IS-LM-FX diagram, show the effects on Y , i , $E_{h/f}$, C , I , and TB .

A fall in autonomous investment reduces aggregate demand at every interest rate, shifting the IS curve **left**.



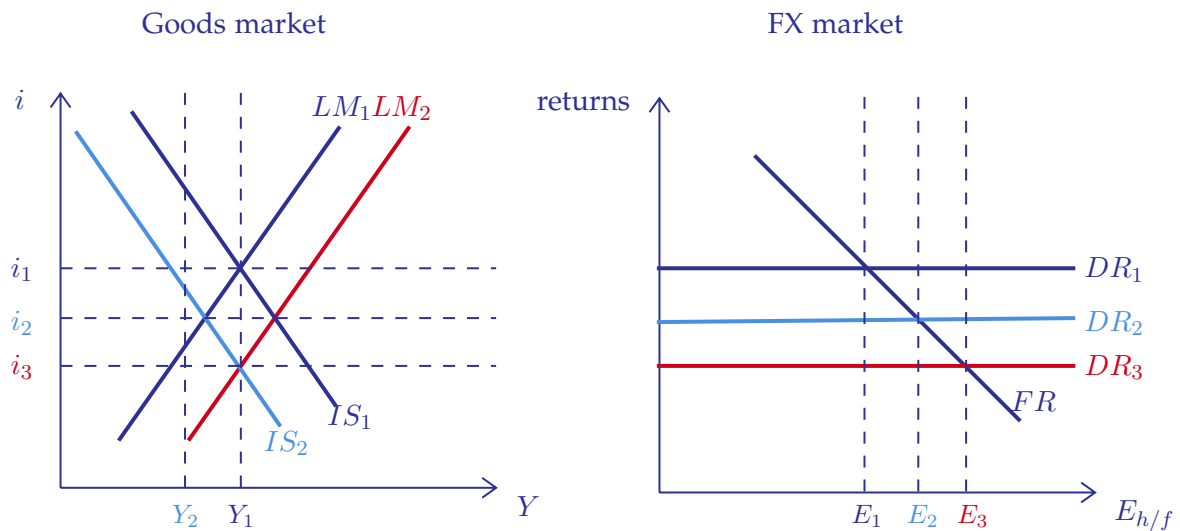
Effects:

- $Y \downarrow$: IS shifts left, output falls
- $i \downarrow$: interest rate falls (movement along LM)
- $E_{h/f} \uparrow$: lower i shifts DR down, currency **depreciates**
- $C \downarrow$: falls with income
- $I \downarrow$: falls from both the autonomous drop and the lower interest rate partially offsetting — net effect is still negative since the autonomous fall dominates
- $TB \uparrow$: depreciation improves competitiveness; lower income also reduces imports. Both channels improve the trade balance.

The depreciation provides automatic stabilization, partially cushioning the output fall, but cannot fully offset it. Note that the trade balance improvement is unambiguous here: unlike a demand shock where income and exchange rate effects can work in opposite directions, both channels move TB in the same direction.

- (b) Now suppose the central bank wants to **stabilize output** at its original level Y_1 using monetary policy. What must it do? Show the effects on Y , i , $E_{h/f}$, and TB relative to the no-policy case in part (a).

To restore output to Y_1 , the central bank must expand the money supply ($M \uparrow$), shifting LM **right**. This lowers the interest rate further and causes additional depreciation, stimulating the trade balance and investment enough to offset the original investment shock.



Effects of monetary stabilization (red), relative to no-policy case (blue):

- Y : restored to Y_1 (stabilization achieved)
- $i \downarrow\downarrow$: falls further to $i_3 < i_2$
- $E_{h/f} \uparrow\uparrow$: depreciates further to $E_3 > E_2$ (DR shifts down further to DR_3)
- $TB \uparrow\uparrow$: improves further relative to the no-policy case, as the larger depreciation boosts competitiveness

Monetary policy successfully stabilizes output by working *with* the automatic depreciation rather than against it. The trade-off is a lower interest rate and a more depreciated currency than in the original equilibrium.

- (c) Instead of monetary policy, suppose policymakers use **fiscal policy** ($G \uparrow$) to stabilize output. Compare the effects on i , $E_{h/f}$, and TB to the monetary policy response in part (b). Which policy is more desirable if policymakers are also concerned about the current account deficit?

Fiscal expansion shifts IS back right, restoring output to Y_1 . Unlike monetary policy, which lowers i and depreciates the currency, fiscal expansion *raises* i and causes *appreciation*.

	After shock (no response)	Monetary policy ($M \uparrow$)	Fiscal policy ($G \uparrow$)
Y	$Y_2 < Y_1$	Y_1 (restored)	Y_1 (restored)
i	$i_2 < i_1$	$i_3 < i_2$ (falls further)	i_1 (returns to original)
$E_{h/f}$	$E_2 > E_1$ (depreciated)	$E_3 > E_2$ (depreciates more)	E_1 (returns to original)
TB	improves	improves further	returns to original

Policy comparison:

- **Monetary policy** stabilizes output while further improving the trade balance through depreciation. It does not crowd out investment (interest rate falls). However, it may be constrained by the zero lower bound if i is already near zero.
- **Fiscal policy** stabilizes output but appreciates the currency, returning TB to its original level with no net improvement. It also raises the interest rate, crowding out private investment.

If policymakers are concerned about the current account: monetary policy is more desirable. The depreciation it induces improves the trade balance, whereas fiscal policy's appreciation offsets the trade balance gains from lower income. This was precisely the tension in the 2001 U.S. policy debate: the Fed favored monetary easing to stimulate growth without worsening the current account deficit, while tax cuts (fiscal expansion) risked appreciation and further external imbalance.

11. (10 points) Real-World Applications

- (a) During the 2008-09 financial crisis, many countries used both fiscal stimulus and monetary easing. Explain why this policy combination makes sense in the IS-LM-FX framework.

The 2008-09 crisis was a massive negative demand shock:

- Consumption fell (wealth effects, confidence), investment collapsed (credit freeze)
- IS shifted far left

Why both policies:

Monetary policy (LM shifts right):

- Lowers interest rates
- Stimulates investment
- Depreciates currency, helping exports
- Fast to implement

But monetary policy may be constrained: Interest rates eventually hit zero lower bound, and thus the quantitative easing has limited effectiveness.

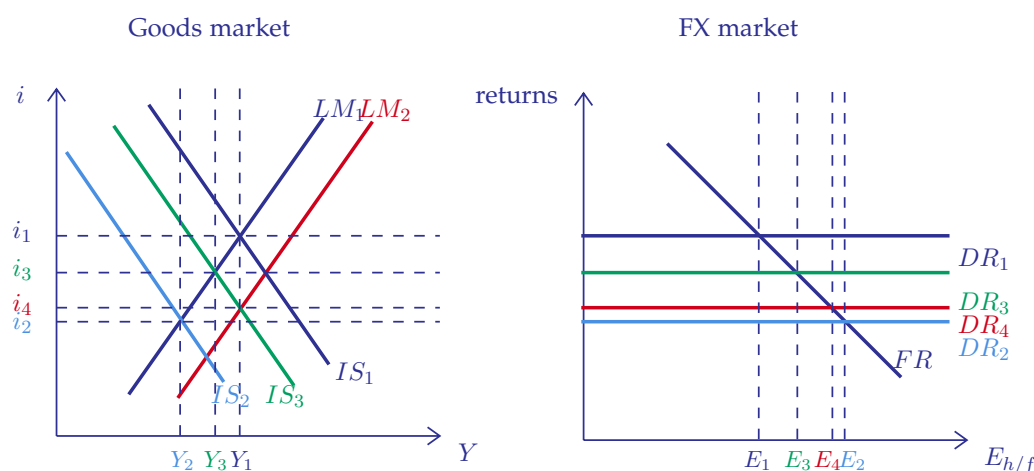
Fiscal policy (IS shifts right):

- More effective at zero lower bound
- No crowding out when $i \approx 0$

Combined effect:

- Fiscal policy expands demand directly but is not enough on its own to restore pre-crisis output
- Monetary accommodation prevents interest rate increases and adds its own expansionary push
- Together they restore output to pre-crisis levels (or near)

We can represent this situation graphically:



The diagram shows four positions. **Black** (IS_1, LM_1): pre-crisis equilibrium at Y_1, i_1, E_1 . **Blue** (IS_2): the crisis shock shifts IS far left, collapsing output to $Y_2 \ll Y_1$ and driving the interest rate down to $i_2 < i_1$, which depreciates the currency to E_2 . **Green** (IS_3): fiscal stimulus shifts IS partially right, recovering output to Y_3 and raising i to i_3 , where $i_2 < i_3 < i_1$ — the interest rate partially recovers but remains below its pre-crisis level, and the exchange rate partially appreciates back to E_3 . Fiscal policy alone is insufficient to restore Y_1 . **Red** (LM_2): monetary easing shifts LM right, adding its own expansionary push. This lowers the interest rate slightly from i_3 to an intermediate level i_4 , where $i_2 < i_4 < i_3$, and depreciates the currency marginally. Crucially, the rightward LM shift raises output further along IS_3 , completing the recovery to Y_1 . Neither policy alone would have been sufficient: fiscal stimulus raised output but left a gap, and monetary easing closes that gap by adding stimulus without pushing interest rates back up.

- (b) Many emerging markets maintain some form of managed exchange rate (dirty float) rather than pure floating or fixed. What might be the advantages of this intermediate regime?

Advantages of managed float (dirty float):

1. Flexibility with stability: This intermediate regimes allows the exchange rate to adjust with fundamentals, smoothing some of the effects of the shocks but still preventing the speculative attacks on the currency to some degree —given that the public is both more tolerant, or less reactive to small depreciations, and aware that the cost of maintaining this

regime is not as severe as with a pure peg.

2. Partial monetary independence: Given that some degree of movements in the exchange rates are allowed, monetary policy can still be used even if with limitations (e.g., cannot move the interest rates to drastically or it would induce strong depreciation pressures).

3. Manage sudden stop episodes: Shocks are managed to slow the depreciations that would follow from a reversal of capital flows, giving firms time to adjust to the new situation and even to mitigate its aftereffects. For a central bank intervening, this still requires a considerable reserves buffer, but it is not as constraining as having to maintain a peg.

4. Accumulate reserves: This regime implies leaning against the appreciation in boom times and building reserves for crisis periods. It provides a type of insurance against the capital flow reversals.

Disadvantages: Sustaining this regime (i) requires large amounts of reserve assets, (ii) can be costly if fighting too strong depreciation market forces, (iii) may lack credibility relative to hard pegs —inviting speculative attacks with a higher probability.

Conclusion: In a nutshell, this type of intermediate regime gives an economy benefits from both ends of the spectrum: it gives some stability to the exchange rate, but permits some degree of monetary independence. However, it does not come without costs and will not give either as much stability as a peg or as much independence as a pure floating regime.

Many emerging markets choose this regime as a pragmatic compromise, especially after experiencing currency crises under fixed rates (e.g., Mexico 1994, East Asia 1997).